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Oefeningenreeks 2E nr. 26.10

$$\log_3(x-3) = \frac{1}{2\log_2 3} + \log_{81}(3x-13)^2$$
$$\Rightarrow \frac{\ln(x-3)}{\ln 3} = \frac{\ln 2}{2\ln 3} + 2\cdot\frac{\ln(3x-13)}{\ln 81} \text{ voor } x > \frac{13}{3}$$
$$\Rightarrow \frac{\ln(x-3)}{\ln 3} = \frac{\ln 2}{2\ln 3} + 2\cdot\frac{\ln(3x-13)}{4\ln 3}$$
$$\Rightarrow \ln(x-3) = \frac{\ln 2}{2} + \frac{1}{2}\cdot\ln(3x-13)$$
$$\Rightarrow 2\ln(x-3) - \ln(3x-13) = \ln 2$$
$$\Rightarrow \ln\left(\frac{(x-3)^2}{3x-13}\right) = \ln 2$$
$$\Rightarrow \frac{(x-3)^2}{3x-13} = 2$$
$$\Rightarrow x^2 - 12x + 35 = 0$$
$$\Rightarrow (x-7)(x-5) = 0$$
$$\Rightarrow \boxed{x \in \{5,7\}}$$

References